

DATA ANALYTICS INTERNSHIP - InternSpark

(AICTE-Listed Internship Program)

REPORT **ON** **WEBSITE TRAFFIC** **ANALYSIS** **(TASK 3)**

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This project analyzes website traffic data to understand user behavior, engagement patterns, and content performance. Key metrics such as sessions, users, bounce rate, and average session duration were calculated to evaluate website effectiveness. The analysis revealed high bounce rates and low user engagement, indicating that users often leave after minimal interaction.

Visualizations identified top-performing landing pages and major traffic sources, while user journey analysis showed limited navigation across the platform. Based on these insights, several recommendations were proposed, including improving landing pages, enhancing navigation, and implementing personalized content strategies.

INTRODUCTION

In today's digital landscape, analyzing website traffic is crucial for understanding user behavior and improving business performance. This project aims to explore traffic logs to identify how users interact with the platform, which pages attract the most attention, and where users drop off.

The analysis helps Alfido Tech optimize its platform for better user experience and higher conversions.

OBJECTIVES

- Analyze user traffic and behavior.
- Identify top landing and exit pages.
- Calculate key metrics (sessions, users, bounce rate).
- Understand user journeys.
- Provide recommendations to improve conversions.

DATASET DESCRIPTION

The dataset contains website interaction logs with the following columns:

- **event** – Type of user action
- **date** – Timestamp of activity
- **country, city** – User location
- **artist, album, track** – Content accessed
- **isrc** – Unique track identifier
- **linkid** – Unique session/page identifier

DATA CLEANING AND PREPROCESSING

The dataset was preprocessed using Python:

- Converted date column to datetime format.
- Removed missing/null values.
- Sorted data chronologically.
- Created **session ID** using linkid.
- Created **user ID** using city and country.

METHODOLOGY

The analysis was performed using:

- **Python (Pandas, NumPy)** for data processing.
- **Matplotlib & Seaborn** for visualization.
- **GroupBy** and aggregation techniques for metrics.

KEY METRICS

- **Total Sessions** – Number of unique session IDs
- **Total Users** – Unique users identified
- **Bounce Rate** – Percentage of sessions with only one interaction
- **Average Session Duration** – Time spent per session

DATA ANALYSIS AND INSIGHTS

- **USER BEHAVIOR**
 - Most sessions consist of very few interactions.
 - Indicates low engagement levels.
- **TOP LANDING PAGES**
 - A small number of tracks dominate user traffic.
 - These act as entry points to the platform.
- **EXIT PAGES**
 - Users tend to leave quickly after initial interaction.
 - Suggests weak content retention.
- **GEOGRAPHIC INSIGHTS**
 - Traffic is concentrated in a few countries.
 - Shows potential target audience regions.

- **USER JOURNEY**

- Users rarely explore multiple pages.
- Navigation flow is limited.

SCREENSHOTS AND VISUALIZATIONS

1.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

In [2]: df = pd.read_csv("traffic.csv")
df.head()
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype  
---  --
0   event       226278 non-null  object 
1   date        226278 non-null  object 
2   country     226267 non-null  object 
3   city        226267 non-null  object 
4   artist      226241 non-null  object 
5   album       226273 non-null  object 
6   track       226273 non-null  object 
7   isrc        219157 non-null  object 
8   linkid      226278 non-null  object 
dtypes: object(9)
memory usage: 15.5+ MB

In [3]: df['date'] = pd.to_datetime(df['date'])
df = df.dropna()
df = df.sort_values(by='date')

In [4]: df['session_id'] = df['linkid']
df['user_id'] = df['city'] + "_" + df['country']
```

2.

```
In [5]: sessions = df['session_id'].nunique()

In [6]: users = df['user_id'].nunique()

In [7]: session_counts = df.groupby('session_id').size()
bounce_sessions = session_counts[session_counts == 1].count()

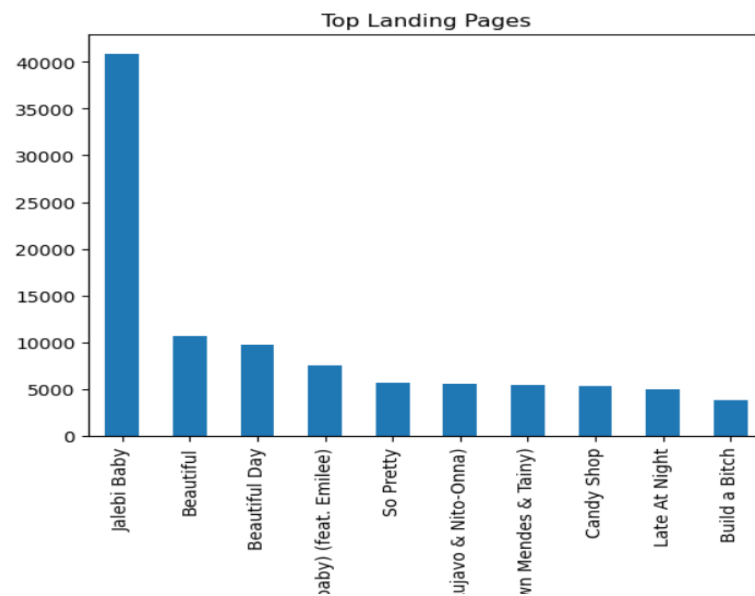
bounce_rate = (bounce_sessions / sessions) * 100

In [8]: session_time = df.groupby('session_id')['date'].agg(['min', 'max'])
session_time['duration'] = (session_time['max'] - session_time['min']).dt.seconds

avg_session_duration = session_time['duration'].mean()
```

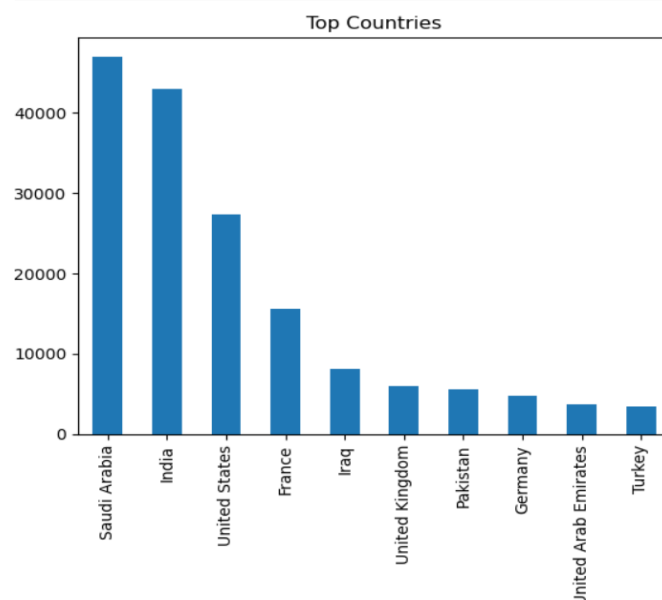
3.

```
In [9]: top_tracks = df['track'].value_counts().head(10)
top_tracks.plot(kind='bar', title="Top Landing Pages")
plt.show()
```



4.

```
In [10]: df['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top Countries")
plt.show()
```



5.

```
In [11]: journey = df.groupby('session_id')['track'].apply(list)
journey.head()

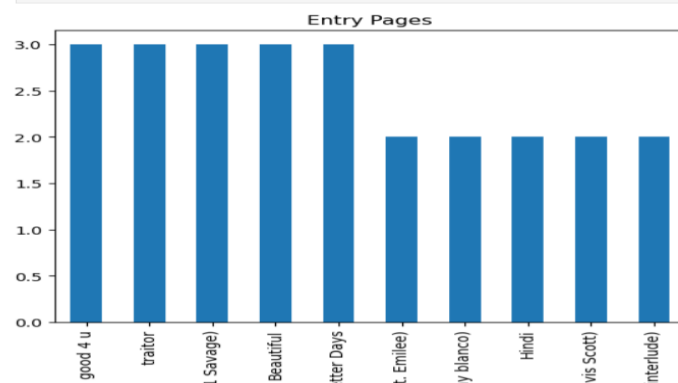
Out[11]: session_id
006af6a0-1f0d-4b0c-93bf-756af9071c06 [dear katara, dear katara, dear katara, dear k...
00759b81-3f04-4a61-b934-f8fb3185f4a0 [Primero Yo, Primero Yo, Primero Yo, Primero Y...
00829040-ee01-4409-966d-d67c7965144a [Beautiful Soul, Beautiful Soul, Beautiful Sou...
009193ee-c3df-4efa-88f2-feb37c0bdfd2 [Ain't It Fun, Ain't It Fun, Ain't It Fun, Ain...
00de7566-f014-4d20-8616-82e4dea45b88 [Wiggle feat Snoop Dogg, Wiggle feat Snoop Dog...
Name: track, dtype: object
```

6.

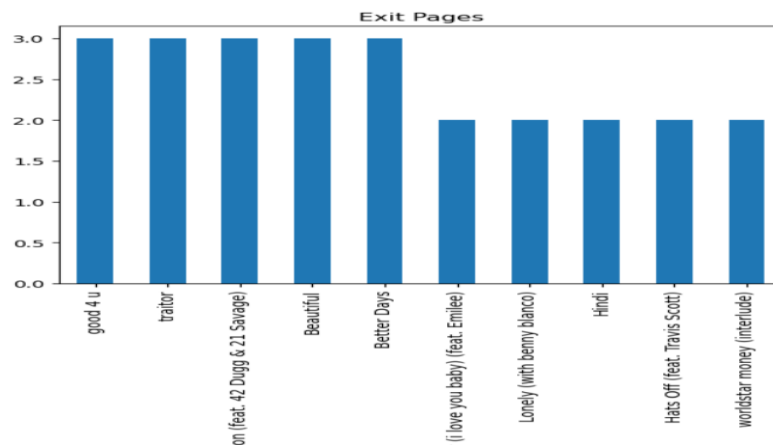
```
In [12]: entry_pages = df.groupby('session_id').first()['track']
exit_pages = df.groupby('session_id').last()['track']

entry_pages.value_counts().head(10).plot(kind='bar', title="Entry Pages")
plt.show()

exit_pages.value_counts().head(10).plot(kind='bar', title="Exit Pages")
plt.show()
```

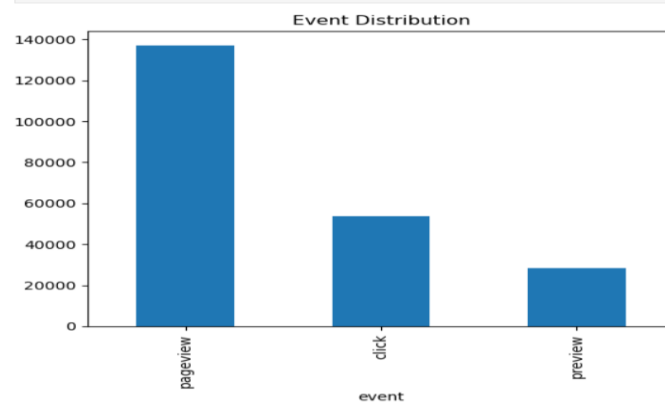


7.



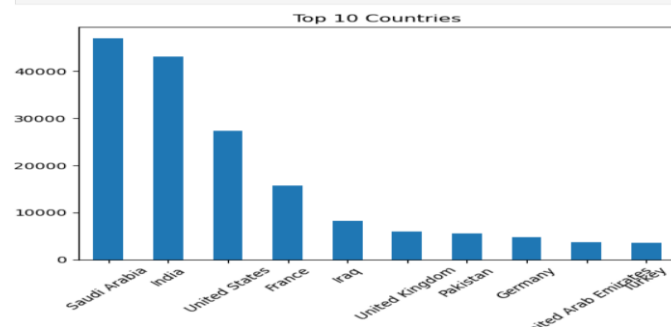
8.

```
In [14]: df['event'].value_counts().plot(kind='bar')
plt.title("Event Distribution")
plt.show()
```



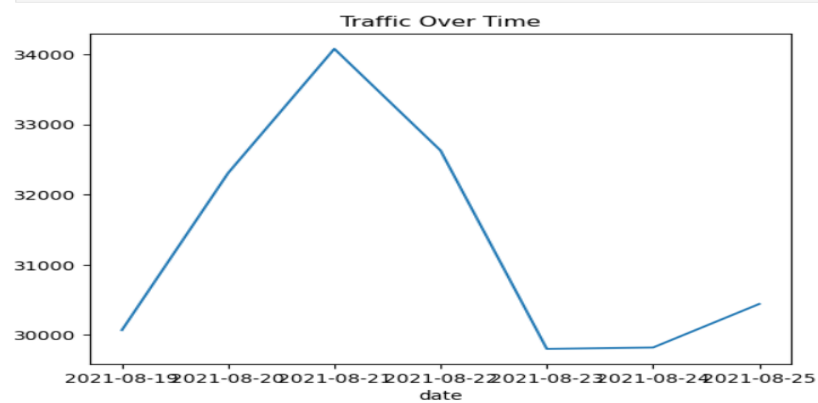
9.

```
In [16]: top_countries = df['country'].value_counts().head(10)
top_countries.plot(kind='bar')
plt.xticks(rotation=45)
plt.title("Top 10 Countries")
plt.show()
```



10.

```
In [19]: df.groupby(df['date'].dt.date).size().plot()
plt.title("Traffic Over Time")
plt.show()
```



VISUALIZATION INSIGHTS : Initial graphs were cluttered due to many categories, so only top values were visualized for clarity.

- Bar charts showed **top countries and tracks**.
- Entry and exit page analysis highlighted **drop-off points**.
- User journey mapping showed **limited navigation depth**.

OPTIMIZATIONS TO IMPROVE CONVERSIONS FOR ALFIDO TECH

1. **Improve Landing Page Experience**

The analysis shows that most users drop off after the first interaction, indicating weak first impressions. Alfido Tech should optimize landing pages by improving UI/UX design, adding clear call-to-action (CTA) buttons, and ensuring faster page load speeds. A well-structured and engaging landing page can significantly increase user retention and conversions.

2. **Reduce Bounce Rate with Engaging Content**

A high bounce rate suggests users are not finding content compelling enough to continue. Introducing personalized recommendations, trending tracks, and interactive elements (such as previews or autoplay features) can encourage users to stay longer and explore more content.

3. **Strengthen Internal Navigation**

User journey analysis indicates that users rarely move beyond one or two pages. Adding features like “Related Tracks,” “Recommended for You,” and easy navigation menus can guide users through the platform, increasing session duration and improving the chances of conversion.

4. **Leverage Geo-Based Personalization**

Since traffic is concentrated in specific countries and cities, Alfido Tech can improve conversions by tailoring content based on user location. Region-specific recommendations, localized promotions, and language customization can make the platform more relevant and engaging for users.

5. **Optimize Conversion Funnel**

Users are likely dropping off before completing desired actions. Simplifying the conversion process by reducing the number of steps, improving page responsiveness, and clearly highlighting key actions (like play, subscribe, or click) can lead to higher conversion rates. Tracking drop-off points in the funnel will also help identify and fix problem areas.

CODE LINK :

https://github.com/shivangikumari1409/InternSpark-Internship/blob/ee6308a912f90b47df04d6e4cc8eddeac8f3ac5f/WebsiteTrafficAnalysis/Website_Traffic_Analysis.ipynb